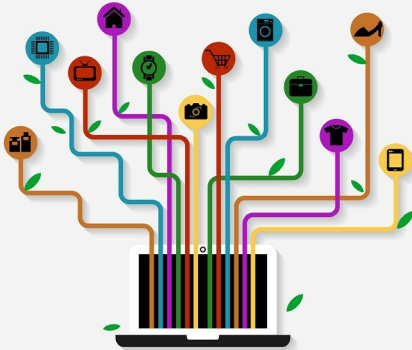


Internet of Things: The Protocols Landscape

The Internet of Things (IoT) extends from a single constrained device to a whole range of cloud systems, all connected by a set of protocols that allow devices and servers to talk to one another. Let's take a look at a few IoT protocols.



The next era of computing is the Internet of Things (IoT), also known as the Internet of Objects. IoT refers to the networked interconnection of everyday objects, which are equipped with ubiquitous intelligence. A recent report by McKinsey Global Institute reported that the number of connected machines has increased by 300 per cent over the past few years. By 2025, the economic impact of IoT is estimated to range from US\$ 2.7 trillion to US\$ 6.2 trillion. Wikibon predicts that the value created from the Internet will be about US\$ 1279 billion in 2020, growing annually at a rate of 14 per cent.

The International Telecommunication Union (ITU) has defined the Internet of Things as, “A global infrastructure for the information society, enabling advanced services by interconnecting (physical and virtual) things based on existing and evolving interoperable information and communication technologies.”

The scope of the Internet of Things is increasing in diverse ways, as IoT based solutions are extending to virtually all areas of everyday life, from smart homes to smart industrial production. And the evolution of Industry 4.0 has begun.

The term 'Internet of Things' consists of two words - Internet

and things. The term 'Things' refers to various IoT devices with unique identities, which have the capabilities to perform remote sensing, actuating and live monitoring of certain sorts of data. IoT devices are also enabled for the live exchange of data with other connected devices and applications, either directly or indirectly, or collecting data from other devices, processing it and sending it to various servers. The other term, 'Internet', is defined as a global communication network connecting trillions of computers across the planet, enabling the sharing of information.

IoT architecture

As the IoT is capable of connecting billions of heterogeneous objects via the Internet, there is an emerging requirement for a dynamic layered architecture. Figure 1 represents a standard IoT layered architecture.

Objects layer: The first layer (perception layer) represents physical sensors of IoT to sense, collect and process the information.

Object abstraction layer: This transfers the data acquired by the object layer to the service management layer via secure channels. Data can be transferred via different technologies like